

CREDIT CARD FRAUD DETECTION

Using Machine Learning to identify suspicious transactions

PROJECT IDEA

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Semester Project • Artificial Intelligence & Data Science

“Detect risk early. Reduce loss. Make digital payments safer.”

1. The Problem We Are Solving

Why fraud detection is a machine learning problem

The real-world challenge

- Millions of legitimate transactions happen every day, while fraudulent ones are a tiny minority.
- Fraudsters change their behavior, making simple fixed rules easier to bypass.
- A model must catch suspicious activity without incorrectly blocking genuine customers.
- The problem is therefore not just classification, it is reliable decision-making under highly imbalanced data.

FRAUD

Rare but high impact

NORMALLY

Large volume of traffic

OUR GOAL

Build a model that flags transactions for review while keeping false alarms under control.

Think of the system as an intelligent “risk filter” placed before the transaction is approved.

2. Where Can This Idea Be Used?

Potential users of a fraud-risk prediction system

Banks & card issuers

Flag suspicious card activity and trigger additional verification.

Payment gateways

Score transaction risk before routing or approving payments.

E-commerce platforms

Identify unusual checkout patterns and account activity.

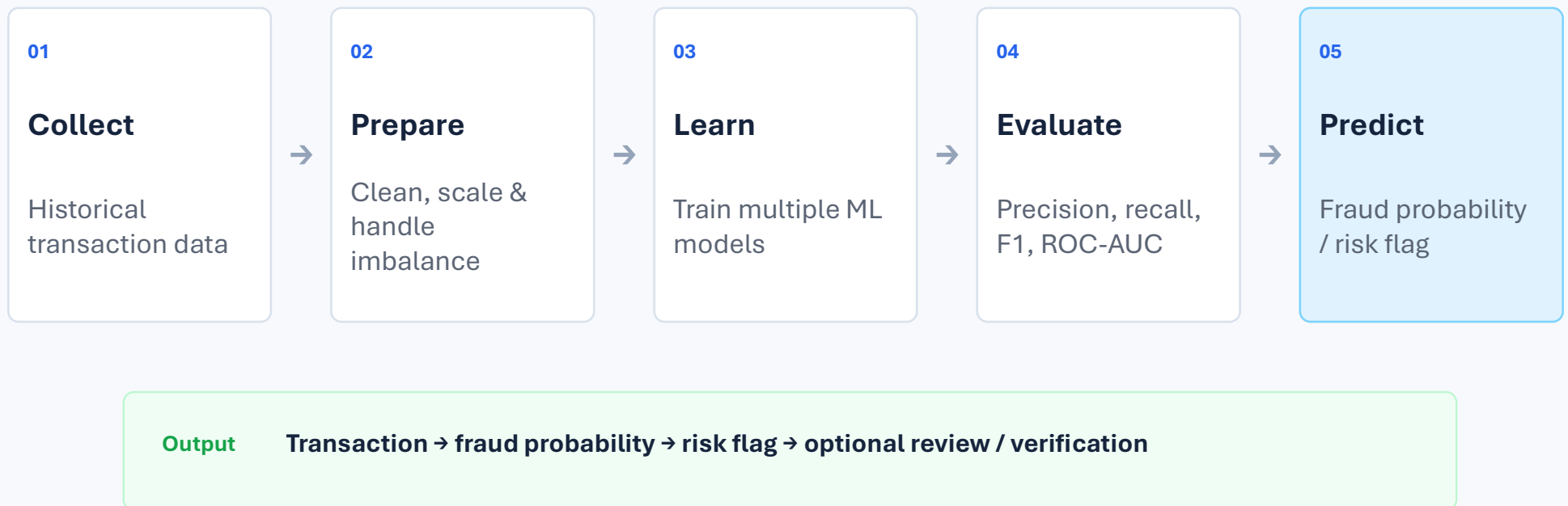
Digital wallets

Monitor transaction behavior and detect abnormal spending patterns.

For our semester project, we will build a prototype decision-support system rather than a live banking approval engine.

3. Our Proposed Solution

An end-to-end ML pipeline



4. What Data Will We Use?

Transaction-level features become the model's evidence

Typical transaction features

Amount How much money was transferred

Time / hour When the transaction happened

Location / merchant signals Where or with whom it occurred

Historical behavior Patterns relative to previous activity

Target label 0 = legitimate • 1 = fraudulent

Important issue: class imbalance

Fraud examples are usually far fewer than genuine transactions.

What we will do

- Scale numerical features
- Use stratified train/test split
- Compare class-weighting / resampling approaches
- Avoid leakage between training and test data

5. Machine Learning Approach

We will compare multiple algorithms rather than trust a single model



Logistic Regression

Strong baseline; easy to interpret



Decision Tree

Captures rule-like patterns



Random Forest

Robust ensemble; handles nonlinear behavior



XGBoost (optional)

High-performance boosting model

Model selection principle

Choose the model that gives the best fraud-detection trade-off - not simply the highest accuracy.

For fraud detection, missing fraud can be costly - so recall and precision matter more than raw accuracy.

6. How Will We Measure Success?

Evaluation designed for an imbalanced classification problem

Precision

Of transactions flagged as fraud, how many are truly fraud?

Recall

Of all actual frauds, how many did we detect?

F1-score

Balance between precision and recall

ROC-AUC

How well the model separates fraud from genuine transactions

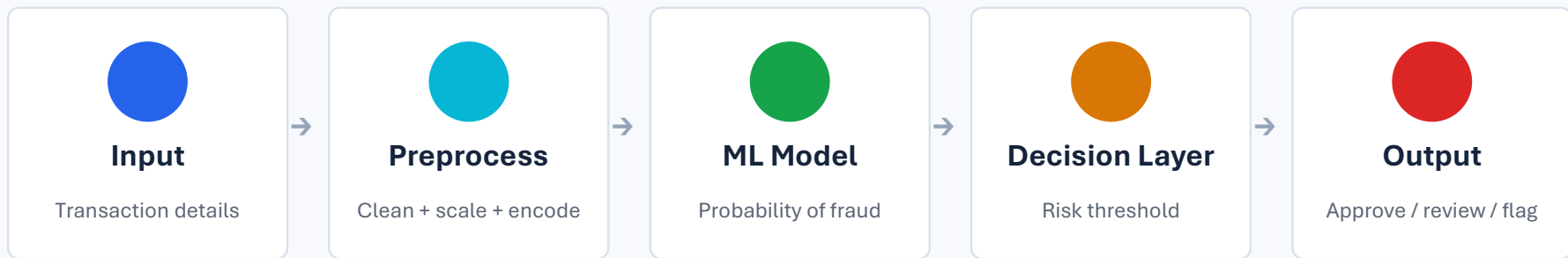
Confusion Matrix

		Predicted	
Actual		TN	FP
		FN	TP

**Goal: maximize true fraud detection
while keeping false alarms
manageable.**

7. System Architecture

How the final prototype will work



Example output Fraud probability = 0.91 → HIGH RISK → send for verification / review

8. How We Will Implement It

Practical technology stack for a semester-scale prototype

Python	Core development	Implementation component
Pandas / NumPy	Data handling & numerical processing	Implementation component
Scikit-learn	Preprocessing, models & evaluation	Implementation component
Matplotlib / Seaborn	EDA & result visualization	Implementation component
Flask / Streamlit	Optional interactive demo	User-facing prototype

Project flow

EDA

Train

Tune

Test

Demo

The prototype can later be extended into an API or dashboard.

OUR PROJECT IN ONE SENTENCE

“We will build an ML-based fraud-risk detection prototype that learns transaction patterns, handles highly imbalanced data, compares multiple algorithms, and flags suspicious transactions for review.”

Problem **Fraud is rare but costly** → **Solution** **ML risk prediction** → **Value** **Safer digital payments**

Thank you

Questions & feedback